

17 Selection and evolution

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Mutation produces new alleles; natural selection changes allele frequencies when variants affect survival and reproductive success in a given environment. Genetic drift can cause chance changes, especially in small populations. Speciation requires reproductive isolation followed by genetic divergence. Selection acts on phenotypes, while populations evolve across generations.

Worked example: follow the reasoning

A pesticide-resistant insect survives treatment and leaves more offspring; the resistance allele becomes more common if heritable, but treatment did not intentionally create the useful mutation.

Quick check · multiple choice

A At the population level, what changes during evolution?

[1]

A Allele frequencies

B Each organism's inherited DNA by need

C Species names only

D Every trait equally

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 In a population exposed to a new drug, explain why resistance increases over generations without claiming individual organisms adapt by need.

[3]

Answers and explanation

Quick check: A. Allele frequencies

Question 1 [3 marks]

Random inherited variants arise before/during exposure [1]; resistant individuals survive and reproduce more [1]; resistant allele frequency rises in the population [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.